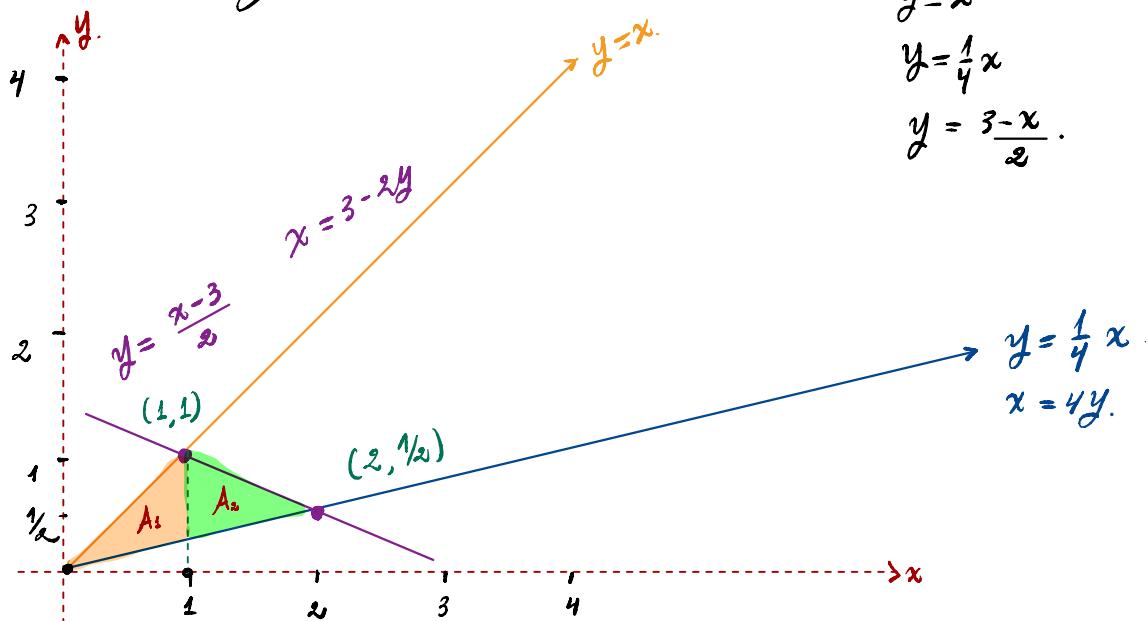


Problem for you: Find the area of the region enclosed by $x+2y=3$, $y=x$ and $y=\frac{1}{4}x$ using.

a). x as variable of integration

b). y as variable of integration.

Sln.



a). x Variable:

$$\begin{aligned}
 A &= A_1 + A_2 = \int_0^1 \underbrace{\left\{x - \frac{1}{4}x\right\}}_{A_1} dx + \int_1^2 \underbrace{\left\{\frac{x-3}{2} - \frac{1}{4}x\right\}}_{A_2} dx \\
 &= \int_0^1 \frac{3}{4}x dx + \int_1^2 \frac{x-6}{2} dx \\
 &= \frac{3}{8}x^2 \Big|_0^1 + \frac{1}{2} \left\{ \frac{x^2}{2} - 6x \Big|_1^2 \right\} = \frac{3}{4}.
 \end{aligned}$$

b). y Variable:

$$A = A_1 + A_2 = \int_0^{1/2} \{4y - y\} dy + \int_{1/2}^1 \{3 - 2y - y\} dy.$$

$$= \int_0^{1/2} 3y dy + \int_{1/2}^1 (3 - 3y) dy.$$

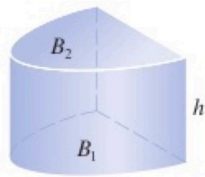
$$\stackrel{\times}{=} \int_0^1 \{3y + 3 - 3y\} dy.$$

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx.$$

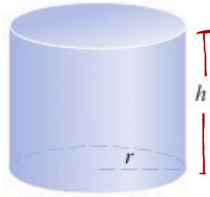
this is wrong! $\triangle$

$$\begin{aligned}
 &= \frac{3}{2} y^2 \Big|_0^{1/2} + \left(3y - \frac{3}{2} y^2 \right) \Big|_{1/2}^1 \\
 &= \frac{3}{8} + \left(3 - \frac{3}{2} - \frac{3}{2} + \frac{3}{8} \right) \\
 &= \frac{3}{4}.
 \end{aligned}$$

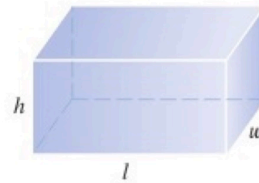
5.2 Volumes We start with a simple type of solid called a cylinder.



(a) Cylinder $V = Ah$

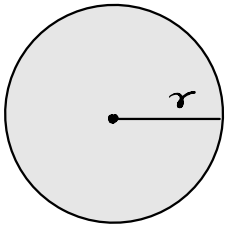


(b) Circular cylinder $V = \pi r^2 h$



(c) Rectangular box $V = lwh$

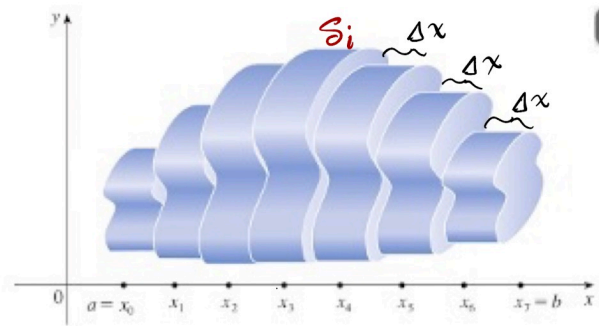
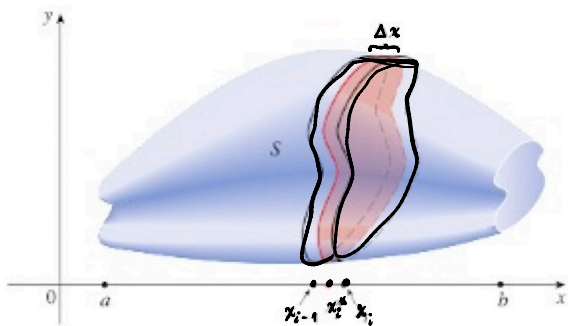
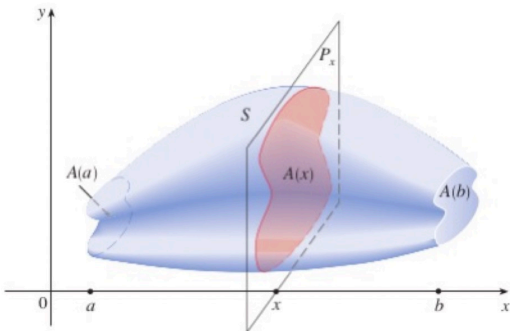
Rmk. Notice that in the case b). $V = \pi r^2 h$ we have a circle in the base.



which area is πr^2 .

We can define the volume $\mathcal{V}$ of a cylinder as $\mathcal{V} = A \cdot h$.

Let's divide S into n "slabs" of equal width Δx by using planes $P_1, P_2, \dots, P_n$ to slice the solid.



The volume of this cylinder is $A(x_i^*) \Delta x$, so an approx to our intuitive conception of the volume of the i th slab S_i is.

$$V(S_i) \approx A(x_i^*) \Delta x$$

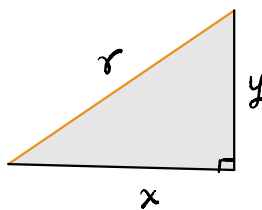
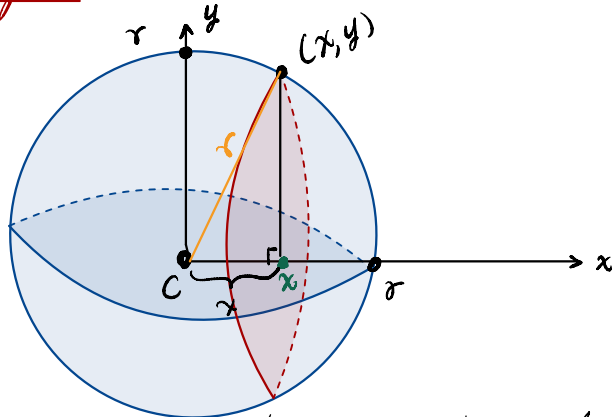
then
$$V \approx \sum_{i=1}^n A(x_i^*) \Delta x.$$

Def. Let S be a solid that lies between $x=a$ and $x=b$. If the cross-sectional area of S in the Plane P_x , through x and perpendicular to the x -axis is $A(x)$, where A is a continuous function, then the volume of S is

$$V = \lim_{n \rightarrow \infty} \sum_{i=1}^n A(x_i^*) \Delta x = \int_a^b A(x) dx.$$

Example. Show that the volume of a sphere of radius r is $V = \frac{4}{3} \pi r^3$.

Sln:



by Pythag. thm
 $y = \sqrt{r^2 - x^2}$

Rmk: Notice that y is the radius of the red circle so the cross-sectional area $A(x) = \pi y^2 = \pi (\sqrt{r^2 - x^2})^2 = \pi (r^2 - x^2)$.

Using the definition of the volume with $a=-r$ and $b=r$

$$V = \int_{-r}^r A(x) dx = \int_{-r}^r \pi (r^2 - x^2) dx$$

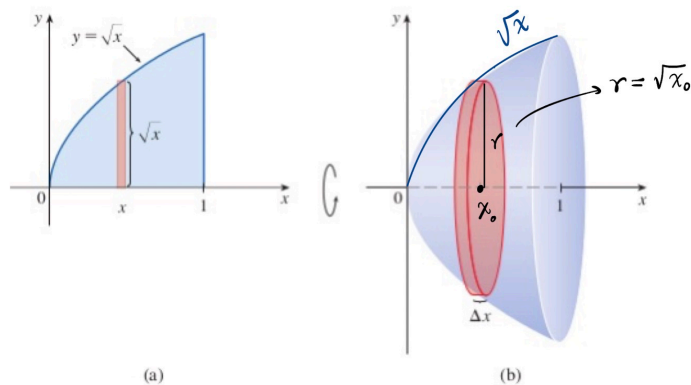
$$= 2\pi \int_0^r (r^2 - x^2) dx. \quad \text{the integral is even.}$$

$$= 2\pi \left(r^2 x - \frac{x^3}{3} \right) \Big|_0^r = 2\pi \left(r^3 - \frac{r^3}{3} \right) = 2\pi \left(\frac{2r^3}{3} \right) = \frac{4}{3} \pi r^3$$

Volumes of solids of Revolution:

Example. Find the volume of the solid obtained by rotating about the x -axis the region under the curve $y = \sqrt{x}$ from 0 to 1.

Sln.



$$A(x_0) = \pi r^2 = \pi (\sqrt{x_0})^2 = \pi x_0.$$

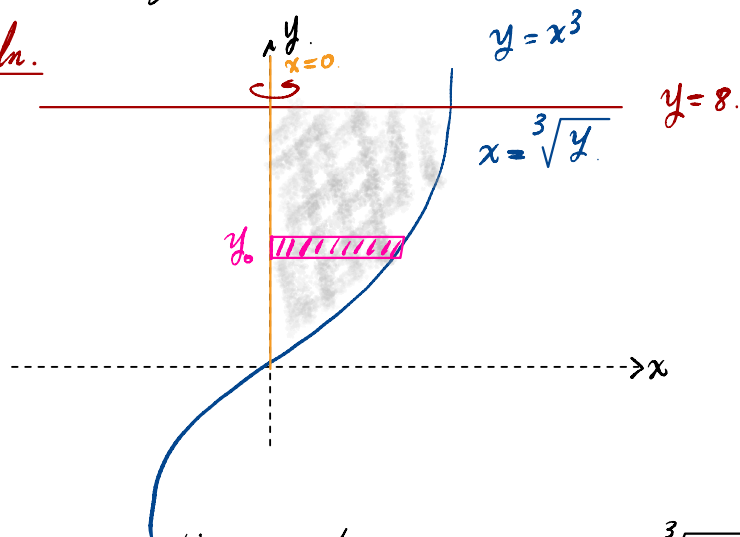
$$A(x) = \pi x.$$

$$V = \int_0^1 A(x) dx = \int_0^1 \pi x dx$$

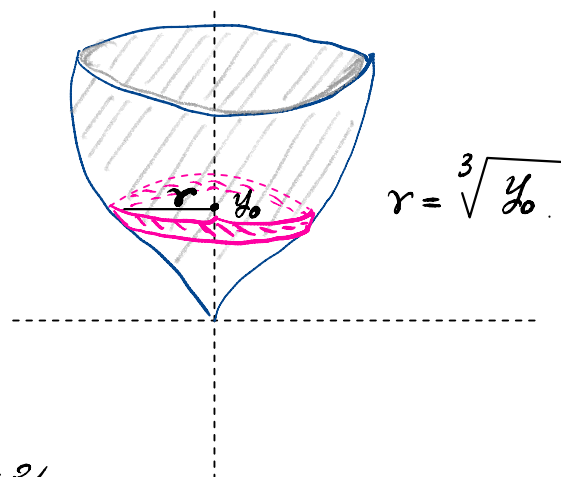
$$V = \left. \frac{\pi}{2} x^2 \right|_0^1 = \frac{\pi}{2}.$$

Example. Find the volume of the solid obtained by rotating the region bounded by $y = x^3$, $y = 8$ and $x = 0$ about the y -axis.

Sln.



becomes



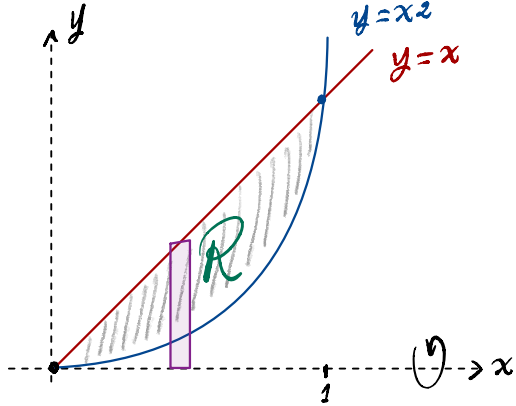
then $A(y_0) = \pi r^2 = \pi (\sqrt[3]{y_0})^2 = \pi y_0^{2/3}$. So,

$A(y) = \pi y^{2/3}$. Using def of volume.

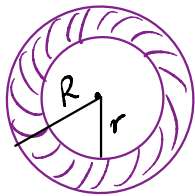
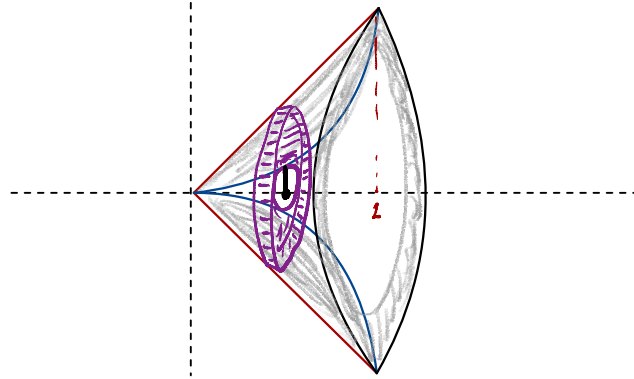
$$V = \int_0^8 A(y) dy = \int_0^8 \pi y^{2/3} dy = \pi \left(\frac{3}{5} y^{5/3} \right) \Big|_0^8 = \frac{96}{5} \pi.$$

Example. The Region R enclosed by the curves $y = x$ and $y = x^2$ is rotated about the x -axis. Find the volume of the resulting solid.

Sln.



Solving $x = x^2 \Rightarrow x(x-1) = 0 \Rightarrow x=0$
 $x=1$.



$$A = A_B - A_o$$

$$r = x^2$$

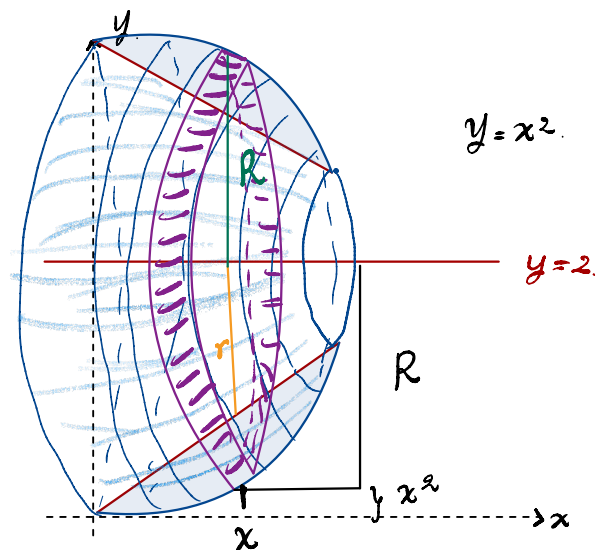
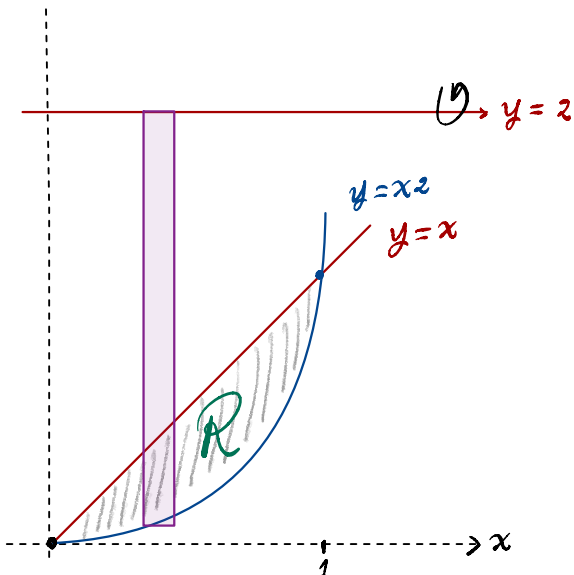
$$R = x$$

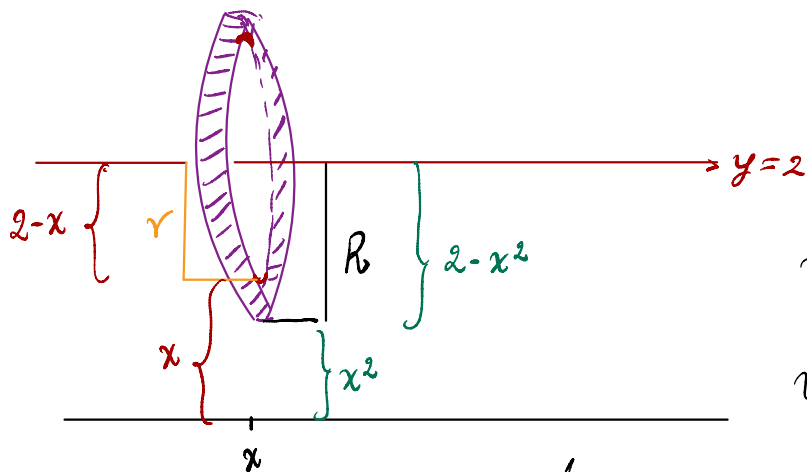
$$A = \pi x^2 - \pi (x^2)^2 = \pi (x^2 - x^4)$$

$$V = \int_0^1 A(x) dx = \int_0^1 \pi (x^2 - x^4) dx = \pi \left(\frac{x^3}{3} - \frac{x^5}{5} \right) \Big|_0^1$$

$$V = \pi \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{2\pi}{15}$$

Example. Find the volume of the solid obtained by rotating the region in the previous example about the line $y=2$.





$$A = \pi(R^2 - r^2).$$

$$A(x) = \pi((2-x^2)^2 - (2-x)^2)$$

$$V = \int_0^1 A(x) dx$$

$$V = \int_0^1 \pi((2-x^2)^2 - (2-x)^2) dx$$

$$V = \pi \int_0^1 \{4 - 4x^2 + x^4 - (4 - 4x + x^2)\} dx$$

$$V = \pi \int_0^1 \{4 - 4x^2 + x^4 - 4 + 4x - x^2\} dx$$

$$V = \pi \int_0^1 \{x^4 - 5x^2 + 4x\} dx$$

$$V = \pi \left\{ \frac{x^5}{5} - \frac{5}{3}x^3 + 2x^2 \right\} \Big|_0^1 = \frac{8\pi}{15}.$$